

Trainings ▾

Preparatory Steps

with Electronically Actuated Injector

⚠ WARNING ⚠

Depending on the circumstances, fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

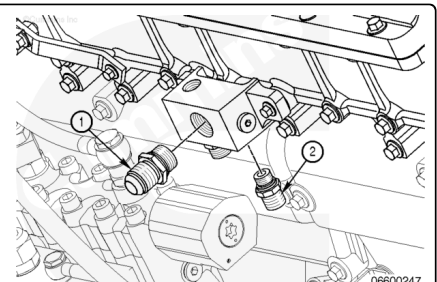
- Clean dirt and debris from the fittings and fuel drain lines.
- Remove the aftercooler assembly. Refer to Procedure 010-002 (</qs3/pubsys2/xml/en/procedures/28/28-010-002.html>).
- Remove the fuel drain lines from the fuel manifold. Refer to Procedure 006-013 (</qs3/pubsys2/xml/en/procedures/28/28-006-013.html>).

Remove

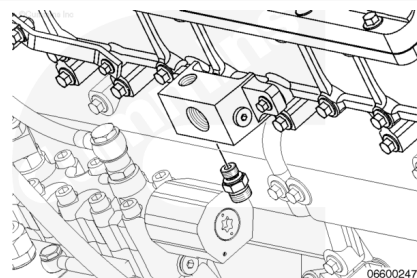
with Electronically Actuated Injector

Remove the two unions from the drain manifold block.

1. This union is used to connect the fuel drain line from the fuel tank to the drain manifold block.
2. This union is used to connect the fuel drain line from the fuel block manifold to the drain manifold block.



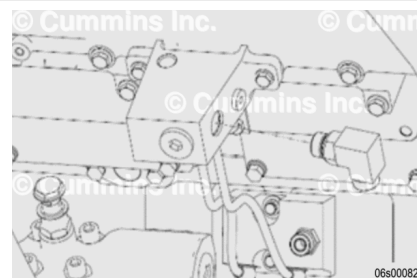
Remove the check valve/union that is used to connect the air bleed line from the fuel pump to the drain manifold block.



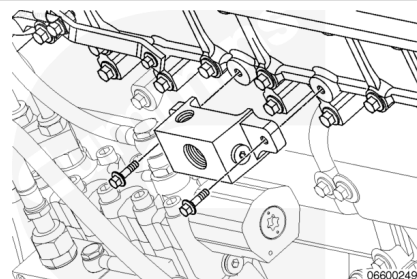
QSK38 and QSK50 Engines with Gerotor Fuel Pump Drain Line

For engines fitted with a gerotor drain line between the drain manifold block and the gerotor, remove the elbow fitting from the side of the block.

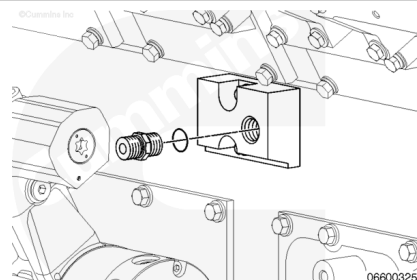
Discard the o-ring.



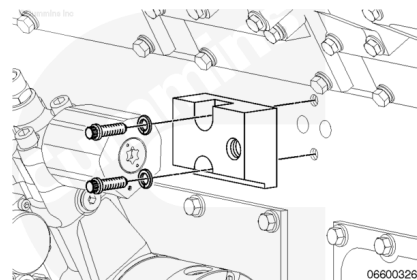
Remove the two capscrews that secure the drain manifold block to the engine and remove the drain manifold block.



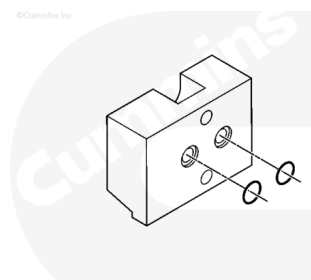
Remove the male union used to connect the fuel drain line from the drain manifold block to the fuel block manifold.



Remove the two capscrews and locking washers that secure the fuel block manifold to the engine.



Remove and discard both o-rings on the rear face of the fuel block manifold.



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Clean and Inspect for Reuse

with Electronically Actuated Injector

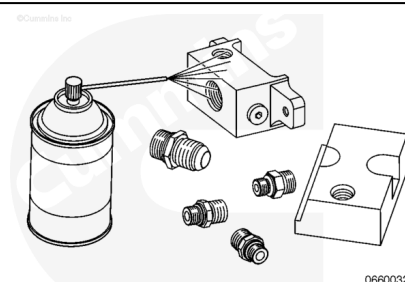
⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

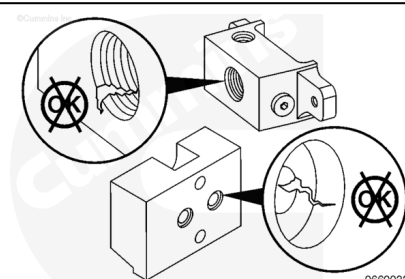
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Clean all parts with contact cleaner, Part Number 4824510, or equivalent.



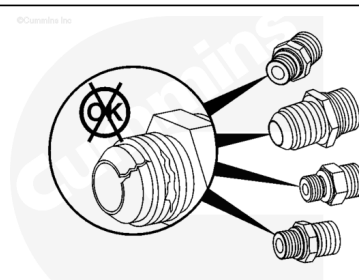
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Inspect the drain manifold block and fuel block manifold for cracks, damage, or stripped threads. Replace if damaged.



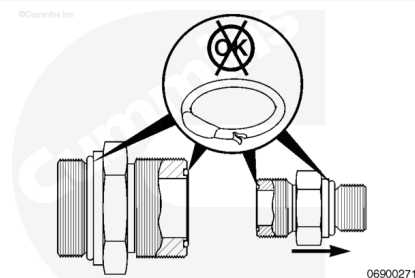
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Inspect the union fitting for cracks, damage, or stripped threads. Replace, if damaged.

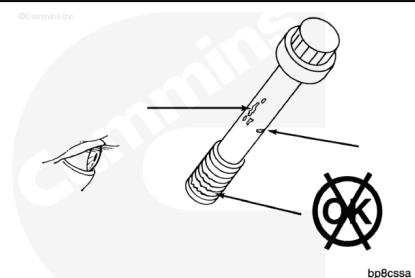


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Inspect the o-rings at each end of the unions/check valve for signs of tears or damage. The o-rings **must** be replaced if damage is found.



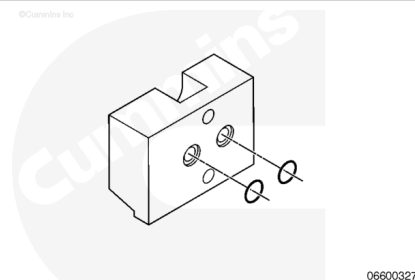
Inspect the capscrews for damage. Replace if necessary.



Install

with Electronically Actuated Injector

Install two new o-rings on the rear face of the fuel block manifold.

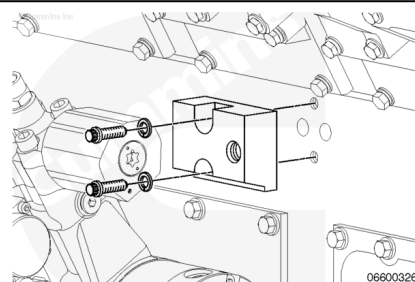


Install the drain manifold block to the engine and tighten the two capscrews.

Torque Value:

Drain Manifold Block Capscrews

1.45 n•m [33 ft-lb]

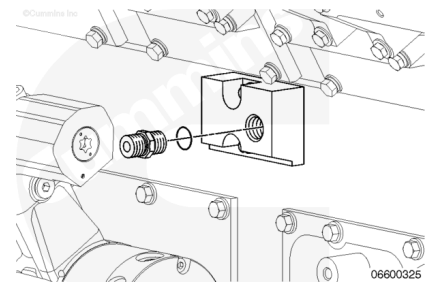


Install the union that is used to secure the fuel drain line from the drain manifold block to the fuel block manifold.

Torque Value:

Fuel Drain Union

1. 70 n•m [52 ft-lb]

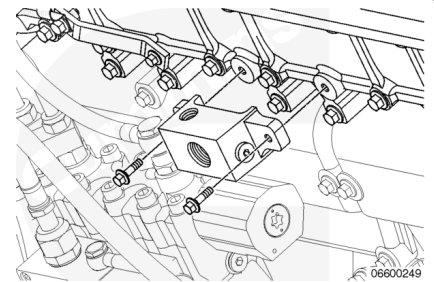


Install the drain manifold block to the engine using the two capscrews.

Torque Value:

Drain Manifold Block Capscrews

1. 20 n•m [177 in-lb]

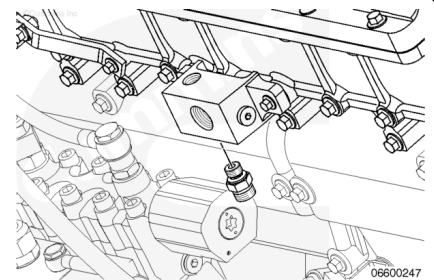


Install the check valve/union that is used to connect the air bleed line from the fuel pump to the drain manifold block.

Torque Value:

Check Valve/Union

1. 55 n•m [41 ft-lb]



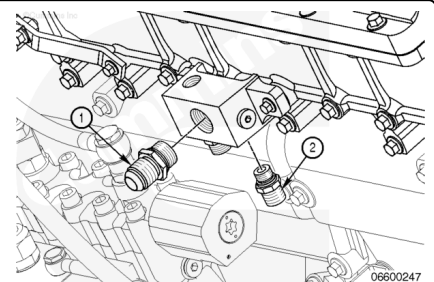
Install the male union (1) that is used to connect the fuel drain line from the fuel tank to the drain manifold block.

Install the male union (2) that is used to connect the fuel drain line from the fuel block manifold to the drain manifold block.

Torque Value:

Fuel Drain Unions

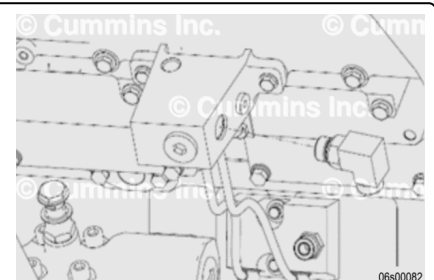
1. 170 n•m [125 ft-lb]



QSK38 and QSK50 Engines with Gerotor Fuel Pump Drain Line

For engines fitted with a gerotor drain line between the drain manifold block and the gerotor, replace the o-ring and install the elbow fitting to the side of the block.

Torque Value: 65 n•m [48 ft-lb]

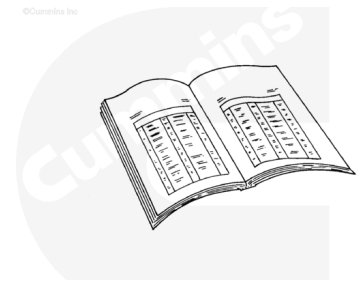


Finishing Steps

with Electronically Actuated Injector

- Install the fuel drain lines. Refer to Procedure 006-013 (</qs3/pubsys2/xml/en/procedures/28/28-006-013.html>).
- Install the aftercooler assembly. Refer to Procedure 010-002 (</qs3/pubsys2/xml/en/procedures/28/28-010-002.html>).
- Operate the engine and check for leaks.

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